



ADS508 Data Science With Cloud Computing

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Company Name: J&F Realty Insighters

Company Industry: Real Estate Analytics

Company Size: 2

Abstract:

J&F Realty Insighters is a real estate analytics company that provides data driven insights into housing market trends. Currently facing the challenge of accurately identifying the factors that influence housing price trends, this project will develop a machine learning model that predicts housing price trends by analyzing housing supply, demand indicators, and historical home values.

Problem Statement:

Real estate investment has traditionally relied on historical trends, expert intuition, and manual data analysis. However, the housing market is a high-stakes environment shaped by a complex web of variables—from hyper-local demographics to broad economic shifts. Without a rigorous, data-driven framework, investors risk inaccurate valuations and missed opportunities in emerging markets.

J&F Realty Insighters bridges this gap. By utilizing machine learning to synthesize massive housing and demographic datasets, we provide precise property value estimations and identify the hidden patterns driving market growth. Our predictive models empower investors to make informed, high-conviction decisions while significantly mitigating financial risk.

Goals:

- **Predict:** Deliver precision property valuations through advanced machine learning.
- **Analyze:** Pinpoint the key demographic and economic factors driving home prices.
- **Scale:** Leverage AWS to build a robust, cloud-native data architecture.
- **Identify:** Help investors discover the next high-growth real estate markets.
- **Optimize:** Prove that data-driven insights outperform manual analysis in property investment.

Non-Goals:

- **Financial Advice:** This project provides data analysis, not personalized investment or financial counseling.
- **Real-Time Data:** Predictions are based on historical datasets rather than live, real-time market feeds.



- Production Deployment: We will not be integrating the model into a commercial-grade real estate platform.
- Proprietary Access: Integration with private or third-party proprietary databases is excluded.
- Short-Term Volatility: Our focus is on long-term trends rather than daily or weekly price fluctuations.

Data Sources [<https://www.zillow.com/research/data/>]:

- Zillow Home Value Index (ZHVI): Represents the estimated typical home value and serves as the primary target variable for our predictive model.
- Median Sale Price: Provides historical transaction data to validate model accuracy against actual market sales.
- Market Temperature Index: Measures supply-demand competitiveness to identify "hot" seller's markets.
- Housing Inventory: Tracks available for-sale stock to quantify the supply-side impact on price fluctuations.

Technical Workflow

The data architecture is built for scale and reproducibility using the AWS ecosystem:

- Storage: Raw historical time-series data is staged in Amazon S3.
- Ingestion: Data is pulled into AWS SageMaker Studio for exploratory analysis and development.
- Transformation: We convert datasets from wide to long format, merging them via shared geographic identifiers (Region/State) to create a unified feature set.
- Modeling: This integrated dataset powers a regression model designed to forecast price trends based on supply and demand dynamics.

Data Exploration:

This project utilizes Amazon S3 for centralized data storage, selected for its scalability, durability, and cost-effectiveness. Using Amazon S3 ensured the integration with the Amazon SageMaker smoothly. The Zillow housing datasets were uploaded via the AWS Management Console to a dedicated bucket, establishing a cloud foundation for the project's dataset. By referencing the data through unique S3 URLs, workflow of manual downloading was eliminated, keeping the workflow within the AWS environment.

- Tools that will be used to ingest and explore data.
 - Amazon S3
 - Central storage location for the datasets. Allows the data to be accessed directly by SageMaker without local file storage.
 - Amazon SageMaker Studio



- Primary development environment for this project. Amazon SageMaker Studio provides a jupyter notebook interface that allows data scientists at J&F Realty Insighters to run Python codes, access S3 data, and perform any necessary data analysis in the cloud.
- Python
 - Python was selected, as it is widely used in data science workflows.

During the data exploration phase, the primary goal is to understand the structure and quality of the datasets before building predictive models. We used commands like `head()`, `shape`, and `columns` to see the structure of the Zillow Home Value Index dataset. Most of the columns represent monthly home value data, while the first few columns contain information about each metro area such as region name and state. We also ran `describe()` to look at summary statistics and used `isnull().sum()` to check for missing values. Overall, the dataset looked fairly complete with only a small number of missing entries.

We then created a simple bar chart to visualize some metro areas with higher home value index values. This helped us quickly see that housing prices vary across regions. At this stage the goal was mainly to understand the dataset and confirm that it could be used to analyze housing trends over time.

Data Ingestion and Exploration Code

The following example code was implemented in a SageMaker Studio notebook to ingest and explore the Zillow dataset stored in Amazon S3

```
In [2]: import pandas as pd

In [3]: df = pd.read_csv("s3://ads508-housing-data-faye/metro_hvi_uc_sfcondo_tier_0_33_0_67_mn_sa_month.csv")
df.head()

Out[3]:
```

	RegionID	SizeRank	RegionName	RegionType	StateName	2000-01-31	2000-02-29	2000-03-31	2000-04-30	2000-05-31	...	2025-04-30	2025-05-31	2025-06-30	2025-07-31	2025-08-31
0	102001	0	United States	country	NaN	120245.401273	120456.951481	120719.304889	121282.001451	121929.597731	...	356520.996883	355847.084537	355240.355926	354820.996430	354597.24283
1	394913	1	New York, NY	msa	NY	216213.074441	217131.858016	218059.151183	219938.205956	221884.032345	...	684527.238403	686280.002733	687742.249805	688904.163124	689687.00471
2	753899	2	Los Angeles, CA	msa	CA	219357.633964	220173.922681	221261.211467	223424.550904	225790.566038	...	943189.059655	938463.706961	934153.202642	931587.011384	930526.40611
3	394463	3	Chicago, IL	msa	IL	149975.869412	150114.703378	150379.115111	151036.905792	151828.148861	...	324141.432142	324425.469135	324778.263087	325598.611099	326571.86483
4	394514	4	Dallas, TX	msa	TX	126453.868825	126510.191820	126574.940868	126743.087346	126964.783873	...	367625.466282	365231.672338	362889.504974	360801.495428	359541.99378

5 rows x 318 columns

Link to the code used for ingestion and exploration: https://github.com/jsryu-git/ADS508_Final/blob/22caef8fb4a463c3028bae6f7dbea27615255b0d/ads508_module3_faye_jun.ipynb

Data Preparation:

- Feature Selection:
 - The feature selection process focuses on variables that influence housing prices based on supply and demand dynamics. Selected predictors include geographic identifiers such as *StateName* and market indicators including housing inventory levels, market temperature index, and the Zillow Home Value Index (ZHVI). Lagged variables and percentage change features are also included to capture

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temporal dependencies. In this study, median sale price is used as the target variable, while ZHVI is treated as a predictor to provide additional information on underlying housing value trends. Metadata fields such as *RegionID* are excluded from modeling but retained for data integration and tracking.

- Feature Creation:
 - New features are engineered to capture both temporal structure and market behavior. Time based variables such as *Year* and *Month* are extracted from the date field to capture seasonality. Lag features including prior month values for ZHVI, inventory, and market temperature are generated to incorporate temporal dependencies. Additionally, percentage change features are computed to reflect market momentum, enabling the model to learn relative changes rather than relying solely on absolute values.
- Feature Transformation:
 - The original datasets are structured in wide format and are transformed into long format using reshaping techniques such as the *melt* function. This transformation allows consistent alignment across time and facilitates merging on region and date. Missing values in predictor variables are handled using forward and backward filling within each region to preserve temporal continuity. Categorical variables such as *StateName* are encoded using one hot encoding within the modeling pipeline. Numerical features are not explicitly scaled due to the use of tree based models, which are not sensitive to feature scaling.
- Data Balancing:
 - Because the task is a regression problem, class imbalance is not applicable. However, the distribution of the target variable is evaluated for skewness. Housing price data often exhibits right skew due to high value markets. While a logarithmic transformation was considered, the model is trained on the original scale to preserve interpretability of predictions.
- Splitting the Dataset:
 - The dataset is split into training and testing sets using an 80 to 20 ratio. In the current implementation, a random split is applied using a fixed random state to ensure reproducibility. While a time based split was initially considered to preserve temporal ordering, the final model uses a random split for simplicity and consistency with standard regression workflows. This approach may introduce minor temporal leakage, which is acknowledged as a limitation.

Link for Module 4: Data Cleaning: https://github.com/jsryu-git/ADS508_Final/blob/a5c592c0d9b0fd2f912697fb0101b4326ce08d47/Module4_DataCleaning.ipynb

Link for Module 4: Modeling: https://github.com/jsryu-git/ADS508_Final/blob/a5c592c0d9b0fd2f912697fb0101b4326ce08d47/Module4_Modeling.ipynb



Model Training:

The model is trained using Amazon SageMaker Studio, which provides a managed notebook environment with direct access to data stored in Amazon S3. The prepared dataset is loaded from S3, and after training, the model artifacts are saved and uploaded back to S3 for reproducibility and future use. This setup ensures simple and scalable cloud based workflow.

The project uses the bring your own script approach with Python and scikit learn. A Random Forest Regressor was selected because its performance for structured tabular data and its capability of capturing nonlinear relationships between housing market variables such as inventory, market temperature, and historical price indicators. Also, Random Forest is robust to outliers and does not require feature scaling, making it well suited for housing data where price distributions may be skewed.

The model is configured with parameters of 200 estimators with max depth of 12 to keep it from overfitting too much while still maintaining its performance. An 80 to 20 train test split was applied, where the model was trained on the training set and evaluated on the test set. Model performance was assessed using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R squared, providing a comprehensive assessment of prediction accuracy and overall model fit.

Link for Module 5: Data Training: https://github.com/jsryu-git/ADS508_Final/blob/main/Module5_Training_JSRYU.ipynb

Measuring Impact:

Two key metrics will be used to evaluate the impact of this project.

Prediction Accuracy (Root Mean Squared Error)

The first metric is Root Mean Squared Error (RMSE), which measures the difference between predicted housing prices and the actual values in the dataset. A lower RMSE indicates that the predictive model is accurately capturing housing price trends.

Model Performance (R^2 Score)

The second metric is the R^2 score, which measures how much variance in housing prices is explained by the model. A higher R^2 score indicates that the model successfully captures the relationship between housing supply, demand indicators, and home values.

Together, these metrics provide a quantitative way to evaluate whether the predictive model improves housing price analysis compared to traditional manual analysis.

Security Checklist, Privacy and Other Risks:

The datasets used in this project do not contain protected health information or personal identifiable information. The Zillow datasets represent aggregated housing market data and do not include records tied to individual homeowners or buyers.

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This project will not track user behavior or store financial information such as credit card data. All datasets are stored in an Amazon S3 bucket and accessed through SageMaker notebooks for analysis.

The application reads data from the following S3 bucket: `s3://ads508-housing-data-faye`

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One potential data bias to consider is uneven geographic representation. Some metropolitan areas may have longer or more complete historical data records than others. This imbalance could cause predictive models to perform better in highly represented regions while underperforming in areas with limited data.

From an ethical perspective, predictive housing models must be interpreted carefully. Housing market predictions could influence investment decisions and potentially contribute to speculation or price inflation in certain areas. Because of this, the model should be presented as an analytical tool rather than financial advice.

Future Enhancements:

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1. The current pipeline uses random 80/20 split for training and testing. Because housing prices are time-series data, a random split can lead to temporal leakage, where the model inadvertently learns from future data points to predict the past. To mitigate this limitation, pipeline could be transitioned to a Time-Series Split. By training on a fixed window of the past and testing on the subsequent chronological period, the model would be forced to learn true predictive patterns rather than just correlating regional trends. Through this transition, RMSE and R Squared can reflect the model's actual utility for investors.
2. Incorporation of external data sources may be a significant enhancement for the model. The current model primarily relies on historical housing indicators such as inventory, median sale price, and lag variables. However, housing prices are also influenced by other economic factors such as interest rates, unemployment rates, regional economic activity, school demographics, and much more. Integrating these external variables could provide a more comprehensive representation of the housing market.
3. Transitioning to AWS Glue for ETL (Extract, Transform, Load) would be a significant improvement for this model. AWS Glue would automate the transformation of Zillow data from wide to long format. This will replace the manual code execution and ensure the new monthly data becomes automatically available in S3. This transition will replace the manual research phase to an automated pipeline phase directly.